

JOSHUA S. SPEAGLE (沈佳士)

Statistical Sciences | Astronomy & Astrophysics

University of Toronto

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MENTORSHIP

I have (co-)mentored/am currently (co-)mentoring **65 individuals**. This includes:

6 postdoctoral researchers

9 doctoral students

2 master's/other students

47 undergraduate students

1 secondary school student

All affiliations outside of the main University of Toronto-St. George (UTSG) campus are listed. The Scarborough and Mississauga campuses are abbreviated as UTSC and UTM, respectively.

Postdoctoral Researchers

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|----|--|-------------------|
| 6. | Ronan Kerr | Fall 2024-Present |
| | Dunlap Postdoctoral Fellow | |
| 5. | Duo Xu | Fall 2024-Present |
| | Co-supervised with Peter Martin | |
| | Schmidt AI & CITA Postdoctoral Fellow | |
| 4. | Biprateep Dey | Fall 2024-Present |
| | Schmidt AI, CITA, & Dunlap Postdoctoral Fellow | |
| 3. | Tanveer Karim | Fall 2023-Present |
| | Co-supervised with Renée Hložek | |
| | Arts & Science Postdoctoral Fellow | |
| 2. | Mike Walmsley | Fall 2023-Present |
| | Dunlap Postdoctoral Fellow | |
| 1. | Jacqueline Antwi-Danso | Fall 2023-Present |
| | Banting Postdoctoral Fellow | |

Doctoral Students

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|----|--|-------------------|
| 9. | Leo Watson | Fall 2024-Present |
| | Co-supervised with Radu Craiu | |
| | Sampling Physical Systems with Quantum Advantage | |

8. Henry Leung Fall 2023-Summer 2024
Primarily supervised by Jo Bovy
Deep Learning Applications to Stellar Spectra
7. Alex Laroche Fall 2022-Present
Co-supervised with Maria Drout
Constraining Binary Stellar Evolution with Middle-Aged Massive Binaries
6. Phil Van-Lane Fall 2022-Present
Co-supervised with Gwen Eadie and Ryan Cloutier (McMaster)
Low-Mass Stars & Exoplanet Population Demographics
5. Alicia Savelli Fall 2022-Present
Primarily supervised by Bart Ripperda and Norm Murray
Evolution of Milky Way Analogues in Cosmological Simulations
4. Mairead Heiger Fall 2022-Present
Co-supervised with Ting Li
Galactic Chemical Evolution with Dwarf Galaxies and Metal-Poor Stars
3. Sam Berek Spring 2022-Present
Co-supervised with Gwen Eadie
Galaxy Globular Clusters Populations with Hurdle Models (with Gwen Eadie)
2. Steffani Grondin Spring 2021-Present
Primarily supervised by Maria Drout and Jeremy Webb (now York)
Stellar and Binary Evolution with Star Clusters
1. Aarya Patil Winter 2021-Spring 2023
Primarily supervised by Gwen Eadie, Jo Bovy, and Sebastian Jaimungal
Asteroseismology with Multitaper Methods

Master's/Other Students

2. Maggie Zhai Fall 2023-Winter 2024
Co-supervised with Gwen Eadie
Statistical Applications in Astronomical Image Analysis
1. Yunyi Shen (UW-Madison, MIT) Summer 2021-Summer 2022
Co-supervised with Vianey Leos Barajas & Gwen Eadie
Modelling Stellar Flares with Hidden Markov Models

Undergraduate Students

47. Alejandro Ortega Cruz Prieto Summer 2024-Present
Co-supervised with Alex Laroche
Building a Multi-Modal Model for Variable Star Spectra and Light Curves
46. Isabelle Laing Summer 2024-Present
Co-supervised with Ting Li and Mairead Heiger

45.

Estimating Stellar Abundances from DESI Spectra with Deep Learning
Matthew Kustec
 Co-supervised with Alicia Savelli

Fall 2023-Present
44.

Exploring Galaxy Assembly Histories Across Cosmological Simulations
Zack Steine
 Co-supervised with Andrew Saydjari

Summer 2024-Fall 2024
43.

Estimating Abundances from Stellar Spectra with Probabilistic Machine Learning
Zack Steine
 Co-supervised with Andrew Saydjari

Summer 2024-Fall 2024
42.

Estimating Abundances from Stellar Spectra with Probabilistic Machine Learning
Michelle Denny
 Co-supervised with Jacqueline Antwi-Danso and Sam Berek

Summer 2024
41.

Counting Massive Galaxies in the Early Universe with JWST
Zehao “Harry” Peng
 Co-supervised with Alicia Savelli

Summer 2024
40.

Exploring Galaxy Evolution in Cosmological Simulations with Machine Learning
Cissy Kuang
 Co-supervised with Alicia Savelli

Summer 2024
39.

Exploring Galaxy Evolution in Cosmological Simulations with Machine Learning
Weizhi “Luke” Shen
 Galactic Paleontology with Probabilistic Machine Learning

Summer 2024
38.

Identifying Strong Lenses in Euclid Data with Active Learning
Khalid Edris
 Co-supervised with Mike Walmsley

Summer 2024
37.

Identifying Strong Lenses in Euclid Data with Active Learning
Junbo “Felix” Li
 Co-supervised with Mike Walmsley

Summer 2024
36.

Identifying Strong Lenses in Euclid Data with Active Learning
Seo Won “Sean” Yi
 Co-supervised with Tanveer Karim

Summer 2024
35.

Assessing the Robustness of Bayesian Model Comparison in Cosmological Analysis
Semoore Tijani
 Co-supervised with Ioana Zelko

Summer 2024
34.

Making the next generation of 3D Interstellar Medium Dust Temperature Maps
Patrice Yee
 Population Demographics of Exoplanets and their Host Stars

Fall 2023-Winter 2024
33.

Population Demographics of Exoplanets and their Host Stars
Jimmy Yang
 Population Demographics of Exoplanets and their Host Stars

Fall 2023-Winter 2024
32.

Population Demographics of Exoplanets and their Host Stars
Xinpeng Xing
 Population Demographics of Exoplanets and their Host Stars

Fall 2023-Winter 2024

31. Wanshi Liang Fall 2023-Winter 2024
Population Demographics of Exoplanets and their Host Stars
30. Yonatan Eyob Fall 2023-Winter 2024
Population Demographics of Exoplanets and their Host Stars
29. Maximilian Zabrodski Fall 2023-Winter 2024
Co-supervised with Steffani Grondin
Multiple Stellar Populations: Observations, Origins, and Onetime Solutions
28. Qianyu “Rita” Fan Fall 2023-Winter 2024
Latent Parameter Inference with Probabilistic Dimensionality Reduction
27. Yihan “Christine” Wang Summer 2023-Winter 2024
Stellar Parameter Estimation with Likelihood-Free Frequentist Inference (LF2I)
26. Belle Pensamiento (George Brown College) Summer 2023-Fall 2023
Co-supervised with Brian Nord (Fermilab)
Classifying PLAsTiCC Transient Simulations with Recurrent Neural Networks
25. Guiping Zhao Summer 2023
Deriving Stellar Properties with Spectrophotometric Modelling
24. Shu Yan “Alice” Wang Summer 2023
Co-supervised with Sam Berek and Gwen Eadie
Applications of Bayesian Predictive Model Comparison in Astronomy
23. Stephanie Lee Summer 2023
Co-supervised with Ting Li and Mairead Heiger
Deriving Stellar Abundances in DESI using Machine Learning
22. Eshal Arshad Summer 2023
Co-supervised with Alicia Savelli
Exploring Galaxy Assembly Histories Across Cosmological Simulations
21. Ishika Bangari Spring 2023
Co-supervised with Alicia Savelli
Exploring Definitions of Milky Way Analogues with Cosmological Simulations
20. Ryan Wang Fall 2022-Summer 2023
Co-supervised with Jeremy Webb (now York) and Steffani Grondin
Identifying Extra-Tidal Stars in APOGEE Globular Clusters
19. Rosalind Liang Fall 2022-Summer 2023
Co-supervised with Jeremy Webb (now York) and Steffani Grondin
Identifying Extra-Tidal Stars in APOGEE Globular Clusters
18. Milica Ivetic Fall 2022-Summer 2023
Co-supervised with Jeremy Webb (now York) and Steffani Grondin
Probabilistic Dimensionality Reduction Methods for Stellar Chemodynamics
17. Ian Chow (now Western) Fall 2022-Summer 2023
Probabilistic Dimensionality Reduction Methods for Stellar Chemodynamics

16. Shahmeer Athar Fall 2022-Spring 2023
 Co-supervised with Bob Abraham
 Rapid Focusing Algorithms for the Dragonfly Telephoto Array
15. Fengqing “Grace” Yu Summer 2022-Summer 2023
 Co-supervised with Ting Li
 Mapping the Milky Way with Blue Horizontal Branch Stars
14. Jinoo Kim (now McMaster) Summer 2022-Summer 2023
 Co-supervised with Lamiya Mowla (now Wellesley) and Kartheik Iyer (now Columbia)
 Estimating Galaxy Sizes from Panchromatic Images
13. Robin Wen (Waterloo, now Caltech) Summer 2022-Summer 2023
 Co-supervised with Jeremy Webb (York University) and Gwen Eadie
 Hierarchical Modelling of Globular Cluster Properties
12. Feiyu Quan (UTSC, now Cambridge) Summer 2022-Spring 2023
 Co-supervised with Neige Frankel
 Investigating Warped Disk Galaxies in IllustrisTNG
11. Charlie Hughes Fall 2021-Spring 2022
 Co-supervised with Ting Li
 Estimating Photometric Metallicities with DECam and S5
10. Eric Conenna (now Kate Spade New York) Fall 2021-Spring 2022
 Co-supervised with Jeremy Webb (now York)
 High-Dimensional Analysis of APOGEE Data
9. Daniel Ding (now Amazon) Fall 2021-Spring 2022
 Co-supervised with Jo Bovy
 Latent Space Decompositions of APOGEE Spectra
8. Ava Oveisi (UTSC) Summer 2021-Fall 2022
 Co-supervised with Kristen Menou
 Imaging Cosmic Dust with Machine Learning
7. Alicia Savelli (Brock, now UTSG) Summer 2021-Summer 2022
 Co-supervised with J. Ted Mackereth (now Defined Benefit Solutions)
 Milky Way Analogues in Cosmological Simulations
6. Jeff Shen (now Princeton) Winter 2020-Summer 2022
 Co-supervised with Neige Frankel
 Estimating Stellar Parameters using 220M Gaia DR3 BP/RP Spectra
 Co-supervised with Neige Frankel and J. Ted Mackereth (now Defined Benefit Solutions)
 Disentangling Stellar Age Estimates from Galactic Chemodynamical Evolution
 Co-supervised with Gwen Eadie, Norm Murray, and Dennis Zaritsky (Arizona)
 Estimating the Mass of the Milky Way with the H3 Survey
5. Mingxuan Teng (now Huawei) Fall 2020-Summer 2022
 Co-supervised with Renée Hložek

- Detecting and Characterizing Outliers in Machine Learning Applications
4. Zhiya Lou (now ByteDance) Fall 2020-Fall 2022
Co-supervised with Gwen Eadie and Jeremy Webb (now York)
Bayesian Model Selection with Globular Clusters
 3. Sina Babaei Zadeh (Western) Summer 2021
Co-supervised with J. Ted Mackereth (now Defined Benefit Solutions) and Lamiya Mowla (now Wellesley)
Studying Galaxy Assembly Histories in Cosmological Simulations
 2. Alan Tu (Harvard) Summer 2020-Summer 2021
Co-supervised with Catherine Zucker (now SAO) and Gus Beane (Harvard)
Kinematics of Young Stars in the Radcliffe Wave
 1. Kaustav Das (IIT Kanpur, now Caltech) Summer 2019-Fall 2020
Co-supervised with Catherine Zucker (now SAO)
Constraining the Distance to the North Polar Spur

Secondary School Students

1. Liam Pilarski (Millburn HS, NJ, now Georgia Tech) Fall 2021-Spring 2022
Estimating Galaxy Properties from Images with Probabilistic Deep Learning